Create data-driven apps faster with ADBC and AI

Who?

Nic Crane

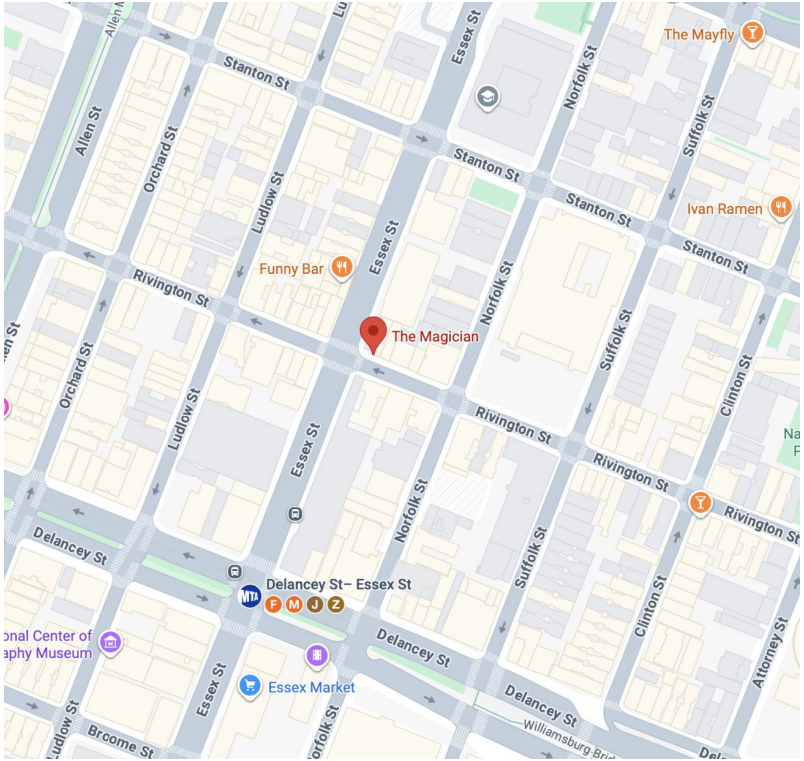
- Arrow PMC Member
- Not an ADBC developer

Columnar crew

- Emil Sadek
- Ian Cook (joining us later)

NYC after-workshop happy hour

Join us tonight after **5:00pm**
at **Magician Bar** on the LES



Housekeeping

💬 Questions in the chat, we'll pause and answer them

💻 Exercises can be found at
github.com/columnar-tech/adbc-lab

🎤 Q&A session at the end with Columnar crew

Assumptions

- Data practitioner
- Comfortable with basic Python code
- Don’t know much about ADBC yet

Polls

- What best describes your role?

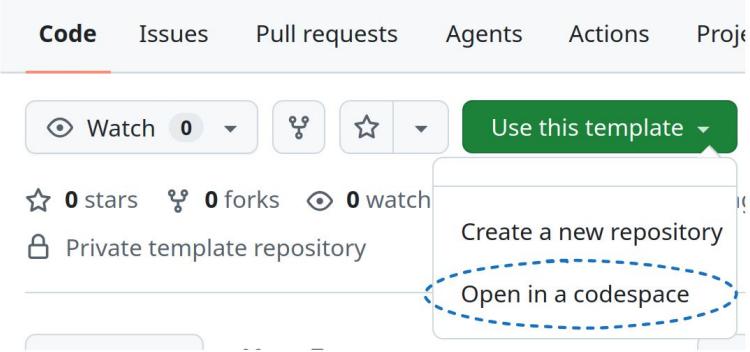
- What's your primary language for data work?

- What data systems or databases do you use regularly?

Exercise Setup

☁️ Option 1: GitHub Codespaces

- Go to github.com/columnar-tech/adbc-lab and click ‘Use this template’ and then 'Open in a codespace'



🖥️ Option 2: Local setup

- Python 3.11+
- uv
- A running PostgreSQL instance

AI coding is changing how we work with data ⚡

We build faster than ever, but the database layer hasn't kept up.

What we'll be learning 🎯

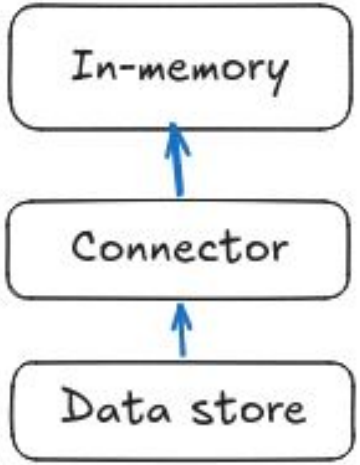
🔧 Use ADBC

🧠 Understand how it works

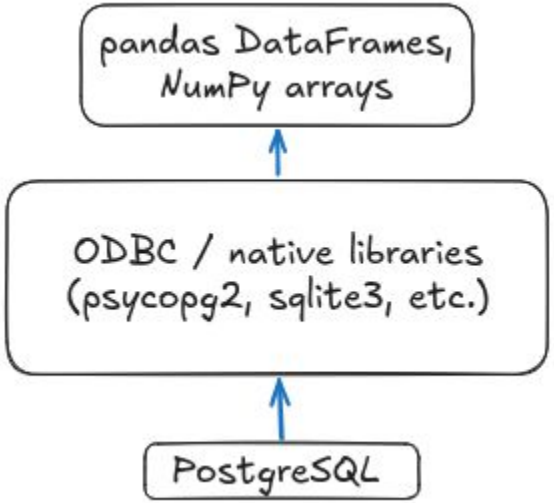
🤖 How it fits it into AI workflows

The Data Connectivity Stack, and Where It's Falling Behind

The data connectivity stack

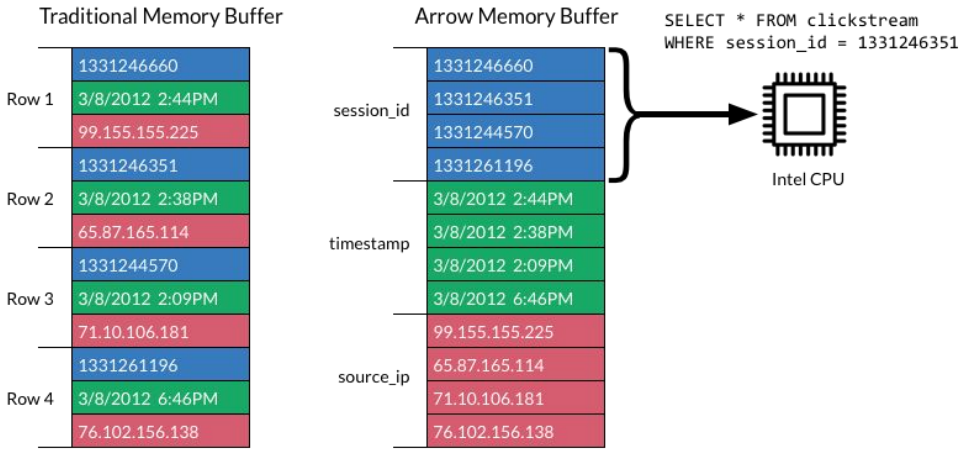


The traditional setup



Columnar data

	session_id	timestamp	source_ip
Row 1	1331246660	3/8/2012 2:44PM	99.155.155.225
Row 2	1331246351	3/8/2012 2:38PM	65.87.165.114
Row 3	1331244570	3/8/2012 2:09PM	71.10.106.181
Row 4	1331261196	3/8/2012 6:46PM	76.102.156.138



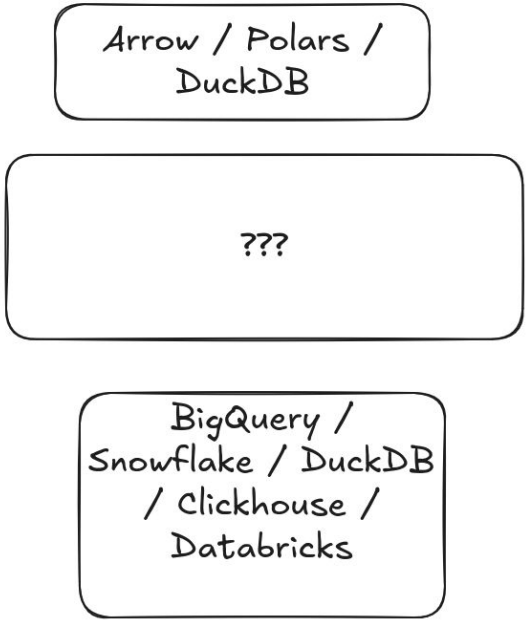
Arrow is everywhere now 🏹

- Apache Arrow: the standard in-memory columnar format
- Polars, DuckDB: built on or closely aligned with Arrow
- pandas 2.0+: Arrow-backed DataFrames available

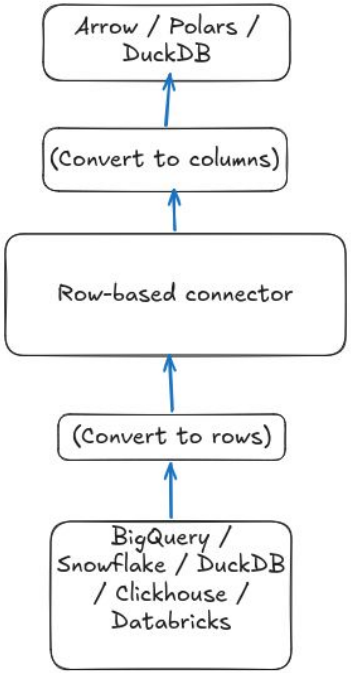
Databases have moved too 🗄️

Columnar	Row-oriented
BigQuery	Postgres
Clickhouse	MySQL
Databricks	SQL Server
DuckDB	
Snowflake	

The mismatch



The mismatch



Demo I: The Speed Difference

Citi Bike trip data

- NYC bike share system, millions of trips
- Each record: start/end station, start/end time, user type, coordinates
- Loaded into a Postgres database
- ~500k rows

Photo: Tom Marvel, CC BY 2.0

Citi Bike Explorer

- Streamlit app

- Typical data analytics use case working on data extract

- Comparing connecting via row-based and column-based connector

Why was it faster? 🤔

Row path: DB → rows → Python converts → DataFrame
↑ slow (Python)

ADBC path: DB → driver converts → Arrow → DataFrame
↑ fast (C/Go/Rust)

✅ Conversion in compiled code, not Python
✅ Arrow arrives natively, no rows touched
✅ No intermediate "bag of tuples" step

When the database is columnar too 🚀

Columnar DB ≈ Arrow format

↓

ADBC driver: almost no conversion needed

↓

Arrow straight through → zero-copy

DuckDB: ODBC vs ADBC 📊

TPC-H lineitem table (SF1), select and insert

No network overhead (client-side database)

ADBC: 38x faster than ODBC 🔥

The entire difference is the connector cost.

Source: duckdb.org/2023/08/04/adbc

Not just Postgres

Backend	ADBC	ODBC (turbodbc)	ODBC (arrow-odbc)	ODBC (pyodbc)	Native	Native Library
BigQuery	1×	2.69×	2.68×	4.17×	1.73×	sqlalchemy-bigquery
DuckDB	1×	driver crashes[1]	driver crashes[1]	1750×+ [2]	1×	duckdb-python

Source: columnar.tech/blog/zero-copy-zero-contest

Questions translate to multiple queries

"Which stations are going to run out of bikes this weekend?"

Question	Query
Where are bikes flowing right now?	SELECT * FROM trips WHERE starttime >= '2026-04-22'
What happened on previous weekends?	SELECT * FROM trips WHERE EXTRACT(DOW FROM starttime) IN (0, 6)
How do stations shift from weekday to weekend?	SELECT * FROM trips WHERE EXTRACT(DOW FROM starttime) IN (5, 6)
Is this week's lead-in worse than usual?	SELECT * FROM trips WHERE EXTRACT(DOW FROM starttime) IN (4, 5)
Which stations have high tourist exposure?	SELECT * FROM trips WHERE usertype = 'Customer'
Score and rank at-risk stations	No new query — combine Steps 1–5

Agentic analytics 🤖

Human: query → **think** → query → **think** → query (you're the bottleneck)

Agent: query → query → query → query → query ... (the connector is the bottleneck)

Lots of round-trips with row-based overhead = real problem

Getting Started with ADBC

What is ADBC? 🏹

- Arrow Database Connectivity

- A modern standard for database connectivity

- Arrow-native: data arrives in the format your tools already use

- ADBC drivers are shared libraries - install once, use from any language

APACHE ARROW
ADBC

Getting set up

1. Install the ADBC driver (database-specific component; pre-built shared library)

2. Install the driver manager (language-specific component; loads the driver and exposes it through familiar APIs)

dbc 🛠️

CLI tool for installing ADBC drivers

One tool, works across all supported databases

dbc

Demo II: Setting Up an ADBC Driver

Basic dbc commands

uv tool install dbc	Install dbc using uv (other methods available)
dbc search	List all available drivers
dbc search -v	List all available drivers (verbose)
dbc info postgresql	Get information about a driver
dbc install postgresql	Install a driver

dbc cheatsheet: docs.columnar.tech/dbc/assets/cheatsheet.pdf

ADBC Python Driver Manager

from adbc_driver_manager import dbapi	Import the DBAPI-compatible interface
dbapi.connect(driver="postgresql", db_kwargs={"uri": uri})	Connect to a database (returns a connection)
conn.cursor()	Create a cursor to run queries
cur.execute("SELECT ... FROM trips")	Execute a SQL query
cur.fetch_arrow_table()	Fetch results as an Arrow table

Exercise 1: Installing dbc & running your first ADBC query

Setup reminder: github.com/columnar-tech/adbc-lab and click ‘Use this template’ and then 'Open in a codespace'.

1. Install dbc

2. Install an ADBC driver for Postgres

3. Connect and run a query

Full instructions in exercise_1.txt

How the pieces fit together ⚙️

Your code → Driver Manager → Driver → Database

```
from adbc_driver_manager import dbapi

dbapi.connect(driver="postgresql", ...)
```

Portability

Scenario: IT said to switch backends 💸

- Snowflake bill was 20K last month
- "Can't you just use <other database> for that workload?"
- Sure, but...

Without ADBC: potentially complicated

- Check type mappings
  - Types can vary between connectors, e.g.
    - decimal → float
    - date → timestamp
    - Null handling
- Update any “glue” code for type conversion
- Driver configuration - ODBC headache or library-specific idiosyncrasies

Each library does it differently, so switching means new surprises

With ADBC: change the driver name ✅

```
# Postgres
dbapi.connect(driver="postgresql", db_kwargs={"uri": "postgresql://..."})

# DuckDB
dbapi.connect(driver="duckdb", db_kwargs={"path": "my_data.duckdb"})

# Same code from here on
cursor = con.cursor()
cursor.execute("SELECT * FROM trips")
table = cursor.fetch_arrow_table()
```

Demo III: Switching Backends

Exercise 2: Switching backends in code

1. **Change the driver from Postgres to DuckDB in your code**
2. **Run the same query against a different backend**
3. **Compare the results**

Full instructions in exercise_2.txt

Portability across languages too 🌍

Python · R · Go · Java · Rust · C/C++ · JavaScript/TypeScript

- same API model
- same Arrow types back

Your data connectivity knowledge transfers, not just your code

One tool across all your databases 🔑

One consistent way to:

- Connect
- Discover tables and schemas
- Query

Why this matters for AI coding 🤖

Native libraries: agent needs psycopg2 API + duckdb API + type converters + ...

ADBC: one API, one set of types, works across backends

Less that can go wrong in generated code

Agents connecting to databases without guidance 🤷

🎭 Hallucinate driver names that don't exist

🎯 Guess at connection URI formats

📚 Fall back to native libraries from training data

The ADBC skill 🧠

Teaches AI coding tools how to use ADBC properly

Connect → discover schemas → query → switch backends

Install: `npx skills add columnar-tech/skills`

https://github.com/columnar-tech/skills

Demo IV: The ADBC Skill

So What Is ADBC and How Does It Work?

ADBC is a standard, like Arrow 📐

	Arrow	ADBC
Standardises	How data is represented in memory	How you connect and get data in and out
Spec first	✅	✅

Definitions

ADBC - the specification; C API is canonical implementation

ADBC Driver - individual library to interact with a database

ADBC Driver Manager - link to this for easier changing of backends

dbc - command line tool for installing ADBC drivers with less effort

What ADBC is and isn't

✅ Cross-platform, cross-language, vendor-neutral, Arrow-native, open

❌ Not a backend, not a wire protocol, not a SQL transpiler, not a replacement for ODBC/JDBC for row-based workloads

How ADBC relates to: 🧩

Technology	Relationship
JDBC/ODBC	Same idea (universal standard), different era. ADBC is the columnar equivalent.
Arrow	Arrow = the data format. ADBC = how to get it in and out of databases.
Flight SQL	A wire protocol. ADBC can use it under the hood, but ADBC is the API you code against.

Where ADBC Fits in Your Workflows

Where ADBC has the most impact ✅

📊 Analytical workloads pulling bulk data

🌍 Multi-language environments needing type consistency

🤖 Agent-driven workflows where simplicity and speed compound

🔄 Anywhere you're converting columnar → rows → columnar

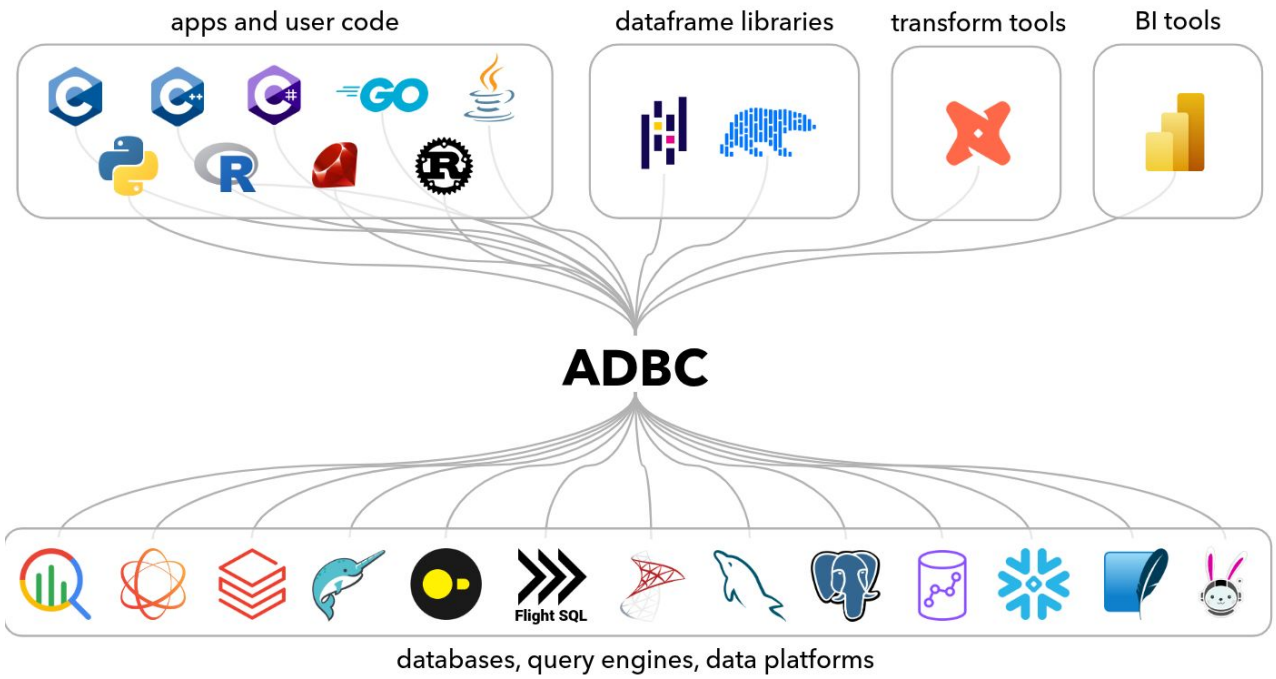
Example: Arrow tables straight into XGBoost, no pandas step (thefulldatastack.substack.com/p/training-xgboost-ducklake-adbc)

When not to use it ⛔

OLTP / row-based transactional workloads (single-row inserts and updates)

ADBC is for bulk columnar access.

You might already be using ADBC



Where to go next 🚀

📖 **Columnar Cookbook** - cookbook.columnar.tech

💻 **adbc-quickstarts** github.com/columnar-tech/adbc-quickstarts

🛠️ **dbc** - columnar.tech/dbc

📝 **Columnar blog** - columnar.tech/blog

📚 **ADBC docs** - arrow.apache.org/adbc

💬 **Columnar Community Slack** - columnar-community.slack.com

ADBC Latest news

ADBC 23 release highlights:

- **ADBC connection profiles**: Now you can store database connection parameters outside your code in a TOML file

- **A new JavaScript/TypeScript driver manager**: Now you can use ADBC drivers in Node.js apps

What's ahead for ADBC 🔮

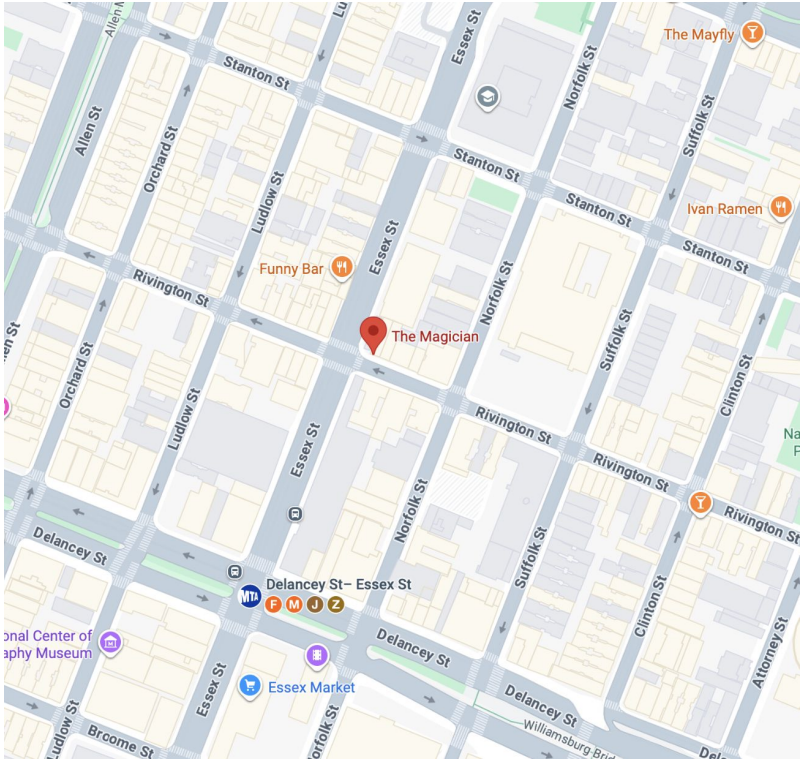
🟢 **Near-term** - Driver Foundry migration (drivers into adbc-drivers), expanded Java support via JNI

🟡 **Mid-term** - ADBC 1.2: more catalog metadata, warnings, requesting specific data types on read

🔵 **Long-term** - Native async APIs for Rust, JavaScript, and other async-first languages

Q&A with Columnar Crew 🎤

Also: please fill in the post-workshop survey!

NYC after-workshop happy hour

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